

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_f9255361-69d\agent-  
a4142c897eec95131.jsonl`  
- SHA-256: `aafbde474486396b6c57d60e2f6d50d87133aff3ba7a1227746c143f57543fe8`  
- Source modified: `2026-06-22T09:35:48+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_f9255361-69d`  
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-22T09:32:46.026Z)

You are auditing ONE doc in the KhelSutra brand/site repo for staleness, as part of a "are all the docs up to date?" sweep.

Repo root: C:\Users\avidu\Projects\khelesutra (bash path: /c/Users/avidu/Projects/khelesutra)
The repo is on `master`, freshly synced to remote HEAD ? the latest merged PR is #94. Today is 2026-06-22.

DOC TO AUDIT: docs/LOCAL_APPLIANCE_ARCHITECTURE.md

Your job: read the doc IN FULL, extract every concrete, falsifiable claim, and VERIFY each against the LIVE repo ? do NOT trust the doc's own wording. Hard discipline rule: a claim is "correct" ONLY when you have file:line or command-output evidence for it; if you cannot verify it, mark verdict "unverifiable" ? never guess, never assume the doc is right. Re-check against actual code/git.

Kinds of claims to check:

- File / path / route references ("served at /capture", "see src/store.js", "public/og.png", "systemd khelesutra-site") -> does the path or route ACTUALLY exist now? Confirm with git ls-files / Read / the routing in site/server.js + site/src/index.js.
- PR numbers and counts ("PRs #1-#8 merged", "20 tests green", "6 languages", "3-step wizard", "P0..P5 done") -> verify against `git -C <root> log`, the actual test output/count, and the code.
- Status markers (DONE / pending / IN PROGRESS / DRAFT / "?" / "owner-gated" / "NEXT") -> still accurate vs the code & git history? Is something marked pending that is actually shipped, or vice-versa?
- Dates, "live at <url>", "merged in #N".
- Whether the doc describes something now SUPERSEDED or fully shipped -> is it an archive candidate (a tracker whose work is complete, or content duplicated by docs/archives/)? Note: docs/archives/FRIENDLY_AND_MULTILINGUAL.md already exists as the archived multilingual tracker.

Tools: Read the doc + every cited file; Grep/Glob to locate code; Bash for `git -C /c/Users/avidu/Projects/khelesutra log --oneline`, `... ls-files`, and greps. To check a test count, you may run the suite (`cd /c/Users/avidu/Projects/khelesutra/site && node --test` or `npm test`), but prefer cheap checks. Verify against master as it stands NOW.

Severity: high = a factually wrong claim a reader would act on (wrong path/route, a "DONE" that isn't, a wrong count/PR that misleads). medium = outdated-but-harmless (stale PR range, old "where we are", superseded framing). low = cosmetic.

Do NOT modify any file ? audit only. Return ONLY the structured object (doc, overallStatus, findings[], summary).

2. assistant (2026-06-22T09:32:48.715Z)

I'll audit the doc against the live repo. Let me start by reading the doc in full and syncing context.

3. user (2026-06-22T09:32:50.159Z)

```
1      # Single-box appliance ? UI-powered rally indexer + reel generator (KhelSutra Studio)
2
3      > **Status:** `DRAFT` ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5
bridge) . **Owner:** Avi . **Created:** 2026-06-20 . **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** $9 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
6      > **Relation:** peer-of `PER_RALLY_SELECTION.md` / `REQUEST_ACCESS_FORM.md` in this
repo; **consumes** the engine's decoupled `StorageBackend` + `python -m backend.worker` seams
and the FastAPI loop; **extends** the engine's `HOSTING_PLAN.md` gating + `UI_PROPOSAL.md` alpha
screens. Engine-side changes here are **proposals to coordinate** with the active
decoupled/worker sessions ? not written unilaterally.
7      > **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine
source this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not
legal/financial advice.
8
9      ---
10
11     ## 0. TL;DR
12     **KhelSutra Studio** turns the engine from a CLI into a **single-box, UI-powered rally
indexer + reel generator**: the khelsutra `web/` React SPA, served behind an **edge auth gate**,
drives the **existing** engine FastAPI loop (ingest ? index ? pick rallies ? reel ? download) on
**one machine**, with `StorageBackend=local` for the MVP. It is the **single-box deployment
profile of the just-shipped decoupled architecture** ? storage=local + compute=local is the
degenerate collapse of the same storage/worker split.
13
14     The **single most important caveat:** the engine has **zero app-layer auth and CORS
`***` `[verified backend/main.py:42]`, so safety is structural **only** because the engine stays
bound to `127.0.0.1:8000` `[verified backend/main.py:638]`, a firewall blocks inbound `8000`,
and a reverse proxy/tunnel is the sole authenticated ingress ? **the gate, not the app, is the
tenancy boundary.**
15
16     Two honest hardware truths: (1) the current droplet is **CPU-only** ? real detection
there is **Gemini (cloud)** or **stub (CPU mock)**; **real local-GPU detection needs a GPU
host** (native WASB, in-flight). (2) The owner priority is to **stand up a GPU host ASAP** ?
provider-agnostic (DO GPU droplet / GCE GPU VM / Cloud Run+GPU) ? but that runs **in parallel**
with the UI MVP, which ships on Gemini today.
17
18     ## 1. Problem & goal
19     The decoupled engine just shipped a `StorageBackend` ABC + a `python -m backend.worker
{infer|train}` dispatcher `[verified backend/storage/, backend/worker/__main__.py]`, and the
FastAPI loop already covers `process ? jobs ? query ? compile ? highlights` `[verified
backend/main.py]`. What's missing is a **console** that lets a non-CLI operator run the whole
loop on a single box, **honestly, behind a gate**.
20
21     **Headline CUJ (owner's words):** a **hosted website** where I pick an input ? **a
Google Drive file, a bucket file (`s3:///`/`gs://`), or a local upload** ? it lands in the
**bucket (DO Spaces / GCS)** or locally ? I **generate highlights** on a **GPU-enabled host**
running the real e2e ? I **pick rallies, watch, and download** ? **completely from the UI.**
22
23     **Outcome ("good")** an operator signs in, picks a source, watches it index, curates
rallies, downloads a reel ? on the **live CPU droplet today** (Gemini/stub), and on a **GPU host
later** (native WASB) with **no UI fork**.
24
25     **Read to get current:** `PER_RALLY_SELECTION.md` ($3/$4/$10 contract anchors), engine
`HOSTING_PLAN.md` (gating), `DECOUPLED_COMPUTE/HANDOFF.md` (what's built),
`VIDEO_LOCALITY_MODEL.md` (privacy/retention).
26
27     ## 2. Decisions locked
```

```

28 | # | Decision | Source / date | Implication |
29 |---|---|---|---|
30 | D1 | Reverse-proxied **single origin**; engine stays `127.0.0.1:8000` | `[verified
main.py:638]`; 2026-06-20 | CORS `` is moot from the browser ? the proxy is the boundary |
31 | D2 | **Edge auth**, zero engine auth code | `HOSTING_PLAN.md`; 2026-06-20 | Caddy
basicauth OR cloudflared + Cloudflare Access |
32 | D3 | MVP drives the **in-process FastAPI worker** loop, `storage=local` | `[verified
job_worker.py:37, local.py:33]`; 2026-06-20 | bucket worker = reserved scale path, out of MVP |
33 | D4 | Embed **RallyPicker** from `PER_RALLY_SELECTION.md` **verbatim** | PR [#11];
2026-06-20 | the console is a superset; do not redesign the picker |
34 | D5 | Console lives in the khelsutra **web/** React scaffold | `web/README.md`;
2026-06-20 | Cloudflare-Pages-style root, not engine static |
35 | D6 | Compute placement is a **setting**, rendered from `GET /api/capabilities` |
`[design]`; 2026-06-20 | UI offers ONLY what the box can honestly do |
36 | D7 | **Ingest is a source menu** (local upload · bucket URI · Drive) normalizing to
one video object | `[verified backend/storage/ref.py parse_uri]`; 2026-06-20 | Drive is a
**stub** today ? scoped honestly (§4.2) |
37 | D8 | GPU host is **provider-agnostic** (DO droplet / GCE VM / Cloud Run+GPU); storage
stays the bucket | Owner 2026-06-20 | compute and storage are separate planes; no provider lock-
in |
38 | D9 | **Observability is first-class** ? structured logs on every new path + a DoD item
| Owner 2026-06-20 | see §5; verified during CUJ replay |
39 | D10 | Delivery = a **self-hosted webserver + UI the user spins up** (NOT a desktop
app); the user points at **local storage + GPU/CPU OR cloud locations** and the UI drives the
tool | Owner 2026-06-20 | **supersedes `DESKTOP_APP_PLAN.md`**; one artifact (this appliance)
runs on the owner box, a droplet, or a GPU host ? same UI, compute/storage are settings |
40
41 ## 3. Foundation ? what the engine gives us today `[verified this session unless
noted]`
42 - **FastAPI loop:** `POST /api/process` (202 + `job_id`) ? poll `GET /api/jobs/{id}` ?
`POST /api/query` ? `POST /api/compile` ? `GET /api/highlights/{video_id}/{filename}`
(FileResponse) `[main.py]`. **Chunked upload exists** ? `POST /api/upload/init` · `PUT
/api/upload/{id}/part/{i}` · `POST /api/upload/{id}/complete` · `DELETE /api/upload/{id}`
`[verified uploads.py:64,89,123,147]` ? but the bundled `app.js` **never calls them** (it only
hits `/api/process` + `/api/compile` `[verified app.js:88,373]`). So browser upload is **client-
wiring only**, no engine change.
43 - **In-process worker:** one module-global `ThreadPoolExecutor(max_workers=1)`
`[verified job_worker.py:37]` ? strictly **one job at a time**; the DB `jobs` table is the
clock; `fail_stale_jobs` sweeps queued/running on boot. Gemini chunk concurrency is a
**separate** knob `indexing.ai_max_workers` (**default 5** `[verified models.py:183]`; set to 1
to serialize) ? it is **not** already 1.
44 - **Storage:** `storage.backend` default = `local` `[verified models.py:298]`;
`LocalStorage.fetch_to_local` is **identity** `[verified local.py:33]`, `put` no-ops on
`src==dst` ? so on a single box the worker's pull?pipeline?push **collapses to read-local/write-
local**. URI scheme `parse_uri` `[verified ref.py]`: bare `local://` ? local; `s3://`
(AWS/R2/**DO Spaces**/MinIO, by endpoint) ; `gcs://`/`gs://` ; `gdrive://` ? **follow-up stub
(emulated only)**. Credentials never in config ? boto3 default chain / `~/.aws` `[verified
s3.py]`.
45 - **Compute seam:** `indexing.detector_impl` ? `{auto=WSL, native=Linux-native,
`stub`=CPU mock}; default `auto` `[verified models.py:129]`, resolved in
`trajectory_hybrid._resolve_runner` `[verified :76-85]`. **Native WASB is in-flight on branch
`feat/native-wasb-runner`** `native_wasb_runner.py` **exists** `[verified]`;
**fusion_hybrid._build_native_runner` returns `NativeWasbRunner`** `[verified
fusion_hybrid.py:86]` while **trajectory_hybrid._build_native_runner` still raises** `[verified
:57-63]`. Acceptance gate = **owner byte-parity** (WSL CSV == native CSV) on a real GPU box ?
not exercisable on the droplet or in the offline suite. Treat native as **not production-
verified**.
46 - **Worker CLI:** `python -m backend.worker {infer|train}` pulls a `--video-uri` ?
pipeline ? pushes `outputs/<video_id>/[intervals.csv,report.json]` `[verified
worker/__main__.py:88,149]`. The validated GPU-free/key-free combo: `detector_impl=stub` +
`--set indexing.skip_ai_handoff=true` + `storage.s3`.
47 - **Gating:** cloudflared Tunnel (outbound-only, no inbound ports) + Cloudflare Access
email-OTP; backend trusts `Cf-Access-Authenticated-User-Email` `[HOSTING_PLAN.md]`. `GET
/api/health` **now EXISTS** (engine M1, [#268]) ? `{status, sport, default_segmenter}`
`[verified main.py:127, corrected 2026-06-21 ? was "does not exist"]`.

```

```

48 - **Droplet reality:** `168.144.126.176`, **CPU-only, NO GPU**, region `blr1`; co-hosts
the marketing site (`/srv/khelsutra-site`, node `:8787`) **and** the engine (`~/rally`); ran the
full train+infer e2e against DO Spaces with the **GPU mocked** `[SETUP_NEW_MACHINE.md,
DECOUPLED_COMPUTE/HANDOFF.md]`. Shared testbed: MinIO `9000/9001` + GCS emulator `9023` also run
there.
49 - **Absent (net-new):** `GET /api/capabilities`, `GET /api/videos` (only single-row
`get_video(id)`), `GET /api/reels/{video_id}`, any source/proxy streaming endpoint, any
**dockerfile**, any per-video **cost ledger** `[verified ? grep returns none]`.
50
51 ## 4. Design / architecture
52
53 ### 4.1 Topology & request path
54 ONE box: **edge gate** (cloudflared+Access or Caddy basicauth) ? **reverse proxy /
static host** serving the `web/` SPA bundle and proxying `/api/*` ? engine **FastAPI
`127.0.0.1:8000`** ? **LocalStorage** ? **detector seam** (Gemini/stub today; native WASB on a
GPU host). The firewall blocks inbound `8000` + the test ports `9000/9001/9023`. The marketing
site stays a **separate** systemd unit on a **separate** subdomain.
55
56 ...
57 PUBLIC INTERNET
58 ? (1) EDGE GATE ? the ONLY ingress; auth lives HERE, not in the app
59 ? cloudflared Tunnel + Cloudflare Access (no inbound ports) OR Caddy :443 +
basicauth
60 ?
61 ?????????????????????????????????????????????????????????????????????????????????
62 ? ONE BOX firewall BLOCKS inbound 8000 + 9000/9001/9023 ?
63 ? (2) reverse proxy / static host (Caddy/cloudflared) ?
64 ? /, /assets/* ? web/ React SPA /api/, /api/highlights/* ? :8000 ?
65 ? ? ONE ORIGIN to the browser ? engine CORS '*' is irrelevant ?
66 ? ? ?
67 ? (3) ENGINE FastAPI (host=127.0.0.1:8000, HARD-bound [main.py:638]) ?
68 ? control: jobs table + ThreadPoolExecutor(max_workers=1) drain ?
69 ? data: uploads.py chunked upload · /api/compile · /api/highlights ?
70 ? ? ?
71 ? (4) STORAGE = LocalStorage (default) fetch=identity · put=no-op(src==dst) ?
72 ? ? ?
73 ? (5) COMPUTE seam: gemini (cloud, key) · stub (CPU mock) [droplet] ?
74 ? native WASB (GPU host) · auto (owner Win+WSL) ?
75 ?????????????????????????????????????????????????????????????????????????????????
76 RESERVED SCALE PATH (out of MVP): console "compute target" selector ? storage=s3 (DO
Spaces)
77 ? SEPARATE GPU host runs `python -m backend.worker infer --video-uri s3://?` ?
outputs/<vid>/ back
78 (needs a NET-NEW orchestration endpoint reflecting outputs into the jobs/poll
contract)
79 ...
80
81 ### 4.2 Ingest sources (the headline) ? one menu, one normalized video object
82 All sources normalize to **"a video the pipeline can read"** ? a local path (MVP) or a
`StorageRef` (scale):
83
84 | Source | Path today | Status |
85 |---|---|---|
86 | **Local upload** | SPA chunks the file ? `max_part_mb` via the existing
`/api/upload/init|part|complete` `[verified uploads.py]` ? server path ? `POST /api/process`.
(Scale: also `StorageBackend.put` to `s3://`/`gcs://`.) | **wire client only** (endpoints exist)
|
87 | **Bucket URI** (`s3://`, `gs://`) | **`POST /api/process` accepts the bucket URI
directly** (resolve ? fetch to scratch ? same pipeline ? API DB) `[verified main.py:159,274,283
? corrected 2026-06-21; was "net-new"]`; the worker CLI also fetches `--video-uri`. So the
**engine endpoint is NOT net-new** ? only the SPA call that POSTs the URI + polls `/api/jobs`
is. *(Default single-user mode; HOSTED mode w/ `allowed_video_root` correctly rejects a bucket
URI 400.)* | **engine: DONE (verified); SPA ingest call: design (= B-P2)** |
88 | **Google Drive** (`gdrive://`) | `parse_uri` recognizes it, but the gdrive backend is
an **emulated stub** `[verified ref.py, DECOUPLED_COMPUTE research]` | **stub ? roadmap; MVP

```

```

requires a Drive?bucket pre-step; do **not** present Drive as live until the backend is real |
89
90     ### 4.3 Compute placement matrix (one UI, surfaced from `/api/capabilities`)
91     | Placement | Runs | Needs | Real/mock | Honest caveat |
92     |---|---|---|---|---|
93     | **CPU droplet ? gemini** (MVP target) | `segmenter=gemini`, in-proc worker |
`GEMINI_API_KEY`; network; **no GPU** | **REAL** | Cloud brain: uploads ?1280px chunks to Google
? **must be in consent copy**; **no per-video cost ceiling** today |
94     | **CPU droplet ? stub** | trajectory hybrid + `detector_impl=stub`+`skip_ai_handoff` |
nothing | **MOCK** | not real tracking ? UI must label "stub (CPU mock ? not real detection)";
demo/regression only |
95     | **CPU droplet ? motion** | `segmenter=motion` | nothing | REAL, low-quality | only no-
key/no-GPU real segmenter; never surface `server/receiver/events` (fabricated) |
96     | **GPU host ? native WASB** (in-flight) | `wasb`+`detector_impl=native` ?
`NativeWasbRunner` (in-proc torch) | a **GPU host ? DO GPU droplet / GCE GPU VM / Cloud
Run+GPU**; `WASB_WEIGHTS` | **REAL**, on-device | partially wired (fusion builds it, trajectory
refuses); **owner byte-parity gate**; **Cloud Run/GCE need a containerized worker ? no
Dockerfile exists, and WASB's torch-1.11 conda env makes the image non-trivial** |
97     | **Owner Win+WSL ? auto** | `wasb`+`detector_impl=auto` (WSL2+conda) | Windows+WSL2+GPU
| REAL (today) | Windows-only transport; the working real-GPU path for owner verification |
98     | **Remote GPU worker via bucket** (reserved) | console CPU + `storage=s3`; separate GPU
host runs `backend.worker infer` | bucket creds; a GPU host; a **net-new orchestration
endpoint** | REAL (remote) | FastAPI loop and worker CLI are **separate** today; "remote GPU as
a setting" is **design** ? selector shown **disabled** in MVP |
99
100     **Provider note:** **Cloud Run + GPU** (scale-to-zero, pay-per-job) is the strongest fit
for the engine's own **per-video-billing** economics (`DECOUPLED_COMPUTE` #246) ? gated on the
**containerization feasibility check** above.
101
102     ### 4.4 Auth / exposure (non-negotiable invariant)
103     No engine auth, by design; the gate is external. **Invariant (all three must hold):**
engine bound to `127.0.0.1` + firewall closes `8000` + edge gate up. Go-live checklist
`[HOSTING_PLAN.md:102-108]`: Access/basicauth ON; **rotate the chat-shared `GEMINI_API_KEY`**;
set `security.allowed_video_root`; rate-limit; per-user quota.
104
105     ### 4.5 UI operator console (web/ SPA ? superset of RallyPicker)
106     - **S1 Ingest & Process** ? the source menu ($4.2) + a segmenter picker from
`/api/capabilities` ? `POST /api/process`.
107     - **S2 Progress** ? 3s poll of the free-text stage
(`hashing/validating/segmenting/finalizing`), a **"one job at a time"** notice, and the restart-
failure terminal state.
108     - **S3 Rally pick ? reel** ? **embed RallyPicker verbatim** (`PER_RALLY_SELECTION.md`):
honor `min_shot_count`; never render motion `server/receiver/events` or a fake confidence meter.
**Per-rally is ONE screen of the console.**
109     - **S4 Reels / History** ? needs `GET /api/videos` + `GET /api/reels/{video_id}` for
**reload-survival**.
110     - **S5 Settings** ? render `config` + `capabilities`; the compute/storage **target
selector shown DISABLED** ("single-box: local") until the bucket-worker bridge lands.
111
112     ### 4.6 Reuse map
113     **Reused (no re-plumb):** the full `/api` loop; the in-process worker; the chunked
upload endpoints (un-called); LocalStorage default; the detector seam incl. the landing
`NativeWasbRunner`; the **RallyPicker** design + anchors; the `web/` scaffold; the
`HOSTING_PLAN` gate; the `SETUP_NEW_MACHINE` runbook. **Net-new:** Caddyfile + systemd units;
`GET /api/capabilities`; `GET /api/videos`; `GET /api/reels/{video_id}`; `GET /api/health`; the
SPA console screens; ingest-from-bucket wiring; a retention sweep; (reserved) the bucket-worker
bridge.
114
115     ## 5. Observability ? logging & monitoring `[required, first-class]`
116     Launch-readiness directive (owner 2026-06-20): **logs and monitoring taken seriously,
with a full log-analysis pass across every new code path.** The engine already uses a
centralized Python logger `[verified main.py:12-17]` ? carry that discipline into every new
path.
117
118     - **Correlation ID end-to-end.** Thread the existing `job_id` (from `/api/process`)

```

****plus**** an ``upload_id`/`request_id`` through ingest ? process ? jobs ? compile ? highlights, and into the worker for the bucket path, so ****one CUJ is traceable**** across UI ? FastAPI ? worker ? GPU host. Log it on every line.

119 - ****Structured logs.**** New endpoints/screens emit structured (JSON-able) logs at ****entry / result / error**** ? not just "it ran". The SPA logs client-side errors + failed fetches with the same correlation id.

120 - ****Loud failures, never silent drops.**** Carry the engine's ethos (#240 "surface failed chunks loudly") + RallyPicker's ``min_shot_count``-warning rule: a partial/empty reel must ****never**** look complete; every dropped clip/chunk is logged + surfaced in the UI.

121 - ****Health + metrics.**** Ship the missing ``GET /api/health`` ``[HOSTING_PLAN.md:42]`` and a thin metrics surface (jobs queued/running/failed, last error, disk free, per-video cost once a ledger exists) for the engine-mode banner + alerting.

122 - ****Aggregation for transient hosts.**** A scale-to-zero Cloud Run / GPU host is ****ephemeral**** ? logs must ship to a durable sink (the bucket or a log service), not die with the instance.

123 - ****Log-analysis pass before merge (DoD item).**** Every PR that adds a code path includes a check that the path logs meaningfully, and the ****logs are inspected during CUJ live-replay**** (§8 Layers 173). Ties `[[feedback-launch-quality-bar]]`.

124

125 **## 6. Risk table**

126 | Risk | Today | Mitigation |

127 |---|---|---|

128 | No app auth + CORS `` `` `[main.py:42]`` | open if any of {localhost-bind, firewall, gate} slips | enforce all three; **\*\*LIVE posture test\*\*** that raw ``:8000`` is refused from outside |

129 | Test ports on shared droplet | MinIO ``9000/9001`` + GCS emu ``9023`` | firewall closes them |

130 | Retention | sweep ****tool**** now exists (`[engine #299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299)`, dry-run-first, uploads >7d, reels durable) but ****not yet auto-scheduled**** (`[engine #301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)`) | run the sweep ``--apply`` before/with any non-owner exposure; auto-schedule per `[#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)`; keep-proxy-only still an option |

131 | Surprise Gemini/GPU bill | ****no cost ceiling/ledger**** | per-video cap; ``ai_max_workers`` (default 5) tunable; ``max_workers=1`` job serialization |

132 | Native WASB unverified | partially wired; owner-parity-gated | treat as not-production-verified until owner byte-parity on a real GPU box |

133 | New engine endpoints | ``capabilities`/`videos`/`reels`/`health`` don't exist | coordinate engine PRs with active sessions; ****branch from `origin/master`, re-verify HEAD, explicit `git add`**** |

134 | Drive ingest overstated | ``gdrive://`` is a stub | don't present Drive as live; require Drive?bucket pre-step until real |

135 | No Dockerfile | runbook-only reproducibility | feasibility-check a worker image before promising Cloud Run/GCE |

136 | Shared-checkout collisions | active worker sessions share the engine tree | the branch-safety protocol on every engine-side PR |

137 | ****Provisioning-credential leak**** | a DO API token / cloud creds could leak if pasted in chat or committed (cf. the ****already-rotated Spaces key**** exposed in chat, ``DECOUPLED_COMPUTE/HANDOFF.md``) | authenticate the CLI ****on-box**** (``doctl auth init`` interactive / ``gcloud`` already authed) so the secret never enters chat or the repo; least-privilege token, revoke after; creds 0600, never committed; ****creating paid GPU infra is a confirm-first action every time**** |

138

139 **## 7. Honest limits ? what this does NOT do**

140 Does ****not**** add multi-tenant identity/RLS (single-tenant alpha; uploads/outputs global). Does ****not**** run real WASB on the CPU droplet (Gemini or stub only). Does ****not**** make the remote-GPU bucket path work (design; selector disabled). Does ****not**** provide in-UI source playback, job cancel, or a true progress bar (coarse free-text stage, poll-only). Does ****not**** containerize (runbook; no Dockerfile ``[verified]``). Does ****not**** verify native WASB (owner byte-parity on a real GPU box, not this offline suite). A green SPA + contract tests prove the ****console plumbing****, not engine rally-detection quality (a separate track).

141

142 **## 8. CUJs & live-test plan** ``[required]``

143 ****CUJs:**** ****CUJ-1**** ingest (Drive/bucket/local) ? index ? reel ? download (core);

```

**CUJ-2** re-open a past match after reload; **CUJ-3** pick-by-query (deterministic filters +
honesty-hiding); **CUJ-4** settings/compute placement (truthful capabilities).
144
145     **Live-test plan ? 5 layers, every CUJ replayed LIVE throughout development:**
146     - **L1 ? Mock engine (hermetic, fast, default + CI).** A tiny FastAPI/MSW implementing
the **exact** contract (upload; `process` 202; `jobs` state machine incl. restart-failure;
`query` with a motion fixture for honesty-hiding; `config`; `capabilities` per box profile ?
CPU-no-key / CPU-with-key / GPU; `compile`; `highlights` tiny mp4; the new `videos`/`reels`).
The SPA drives **all 4 CUJs in <1s**, **replayed via the Claude Preview/browser tools** (start
dev server ? fill the ingest picker ? click through process/poll/RallyPicker ? assert the reel
link via snapshot/network). Catches reload-survival (CUJ-2), the `play_segments` field-name bug
(CUJ-3), capability honesty (CUJ-4).
147     - **L2 ? Engine-contract checks (real FastAPI, GPU-free).** Boot `python -m backend.main
--server` with `storage=local` + `detector_impl=stub` + `skip_ai_handoff` and assert the **mock
and the real engine agree shape-for-shape** (202 keys, jobs enum, `result.play_segments` schema,
compile `{status,filepath}`, highlights content-type). Every new endpoint is born with a mock
handler **and** a real-engine contract test **in the same PR**. Runs in CI.
148     - **L3 ? Real local e2e (the MVP substrate).** Through the gate on the droplet: CUJ-1
with `stub` (key-free) and `gemini` (real, short clip to bound cost); the **exposure-safety
posture test** (raw `:8000`/`:900x` refused from outside) as an assertion, not an assumption;
reuse the `parity.yml` dual-host pattern.
149     - **L4 ? CI regression (green gates merge).** `npm test` + a stub-engine contract job +
headless browser-replay CUJ scripts; one small PR per tracker row off `origin/master`, **no
stacking**; coverage at/above the floor; **logs inspected during replay** ($5).
150     - **L5 ? Owner hardware e2e (handed off).** Native WASB **byte-parity** on a real GPU
box; real-4K **golden-corpus** runs (Gemini + native); Modern-Standby/`cv2`-vs-`ffmpeg`
behavior; a multi-GB browser upload. Reported back into $9, stated plainly as **not verified**
in the offline suite.
151
152     ## 9. Deliverables & progress tracker ? **source of truth**
153     Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking; CI green gates merge; coverage ? floor; each PR
ships its tests + meaningful logs ($5).
154
155     | ID | Deliverable | Depends on | Gated? | Status | PR |
156     |----|-----|-----|-----|-----|----|
157     | T0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | owner | ? |
(this PR) |
158     | T1 | Mock engine + capability fixtures + browser-replay CUJ scripts (L1) | T0 | no | ?
| ? |
159     | T2 | Appliance shell: Caddyfile + systemd units; firewall; **edge auth**; rotate key
(P1) | ? | owner | ? | ? |
160     | T3 | L3 live posture test (raw `:8000`/`:900x` refused from outside) | T2 | no | ? | ?
|
161     | T4 | S1 ingest: **local upload wired ?
([#66](https://github.com/avidullu/khelsutra/pull/66))** ? file picker ? chunked `api/upload`/`
? server path ? `api/process` ? existing poll?load?compile?download; segmenter picker still
pending | T1 | no | ? | [#66](https://github.com/avidullu/khelsutra/pull/66) |
162     | T5 | S2 progress (poll + one-job + restart-failure) | T4 | no | ? | ? |
163     | T6 | S3 embed **RallyPicker** + compile + download | T4 | no | ? | ? |
164     | T7 | *(engine)* `GET /api/capabilities` + `GET /api/health` + banner | T1 | coord | ?
| ? |
165     | T8 | *(engine)* `GET /api/videos` + `GET /api/reels/{video_id}` (S4 reload-survival) |
? | coord | ? | ? |
166     | T9 | S5 Settings (config+capabilities; disabled compute/storage target) | T7,T8 | no |
? | ? |
167     | T10 | Observability pass: correlation id end-to-end + structured logs + metrics ($5) |
T4 | no | ? | ? |
168     | T11 | Retention sweep for `output`/`uploads`/` ? **sweep tool shipped** ([engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299)): dry-run-first; purges
stale uploads >7d + orphan `.partial-*` + regenerable frames/staged/proxies; **reels/DB/CSVs
durable**; `--json` plan ready for a scheduler. Automated scheduling ? [engine
#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301). | ? | owner ?
(uploads 7d, keep reels ? 2026-06-21) | ? | [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) |

```

169 | T12 | Ingest from **bucket URI** (``s3://`gs://``) wiring ? engine accepts URIs
(M2/[#280]) **SPA ingest screen shipped** (B-P2, [#56]); in-UI ``gs://`reel CUJ` proven live
e2e (``DRY_RUN_CUJ.md` $9`) | T4 | ? | ? | engine #280 + SPA #56 |

170 | T13 | **GPU host stand-up** (provider-agnostic) + wire ``detector_impl=native``; owner
byte-parity (P4) | branch | ? owner | ? | ? |

171 | T14 | **(reserved)* Bucket-worker bridge ? remote-GPU "compute as a setting" (P5) |**
T8,T12 | ? owner | ? | ? |

172

173 ****Owner priority (2026-06-20): T13 (GPU host) is elevated to run in parallel from the
start**** ? owner provisions the host + token; the engine session lands native WASB; the UI MVP
does ****not**** block on it (ships on Gemini).

174

175 ****? Alpha deploy path chosen = SPLIT, not the single-origin shell (owner 2026-06-21).****
The alpha ships the ****split**** topology (SPA on Cloudflare Pages + engine on an ``api.``
cloudflared tunnel + CF Access over both ? the engine ``HOSTING_PLAN.md` $2`), ****not**** the single-
origin Caddy shell of ****T2**** (kept as the documented alternative ? `$0/D1`). The ****deploy kit +**
runbook** (``[deploy/DEPLOY_ALPHA.md](../deploy/DEPLOY_ALPHA.md)`` +
``deploy/cloudflared.config.example.yml``) is shipped (? templates+runbook); the actual stand-up
(tunnel + Pages + CF Access + key rotation) is the ****owner**** step. The ****T3****
raw-``:8000``-refused posture test is documented as the deploy kit's \$6 verification. Enabled by:
engine exposure-safety boot posture (engine #294) + SPA ``VITE_API_BASE`` ([#64]).

176

177 **## 10. Open questions ? owner / external**

178 ****Blocking gates:**** (1) ****edge-auth**** ? Caddy basicauth (self-contained) vs
cloudflared+Cloudflare Access (matches the two-host design, no inbound ports)? (2) ****retention**
defaults** ? ****window DECIDED**** (uploads ****7d****, reels durable; [engine
#299](<https://github.com/Khelsutra/badminton-highlight-indexer/pull/299>) tool + scheduling
[#301](<https://github.com/Khelsutra/badminton-highlight-indexer/issues/301>)); ****still open:****
auto-delete originals after compile vs keep proxies+intervals only (privacy/consent). (3) ****per-**
video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4) ****GPU host choice**** ? DO GPU
droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker worth building for Cloud
Run)? (5) ****P5 bridge ownership**** ? who builds the orchestration endpoint + how it coordinates
with the active worker sessions.

179 ****Nice-to-knows:**** console home (web/ vs engine static); a review-playback streaming
endpoint; per-user scoping timing; containerization vs runbook.

180

181 **## 11. Definition of done**

182 - [] CUJ-1 works end to end on the droplet (Gemini ****and**** stub) behind the gate, via
the ingest menu.

183 - [] Raw ``:8000`/`:900x`` unreachable from outside (****tested****, not assumed).

184 - [] Local upload wired; bucket-URI ingest at least designed/tracked; Drive scoped
honestly (not faked).

185 - [] RallyPicker embedded with honesty rules; ``min_shot_count`` surfaced.

186 - [] ``/api/capabilities`` + ``/api/health`` + banner truthful; reload-survival via
``/api/videos`` + ``/api/reels``.

187 - [] ****Observability:**** correlation id threaded end-to-end; structured logs at every
new path; loud failures; logs inspected during CUJ replay.

188 - [] Retention sweep in place (****tool shipped**** [engine
#299](<https://github.com/Khelsutra/badminton-highlight-indexer/pull/299>); ****auto-schedule****
[engine #301](<https://github.com/Khelsutra/badminton-highlight-indexer/issues/301>));
``GEMINI_API_KEY`` rotated; ``allowed_video_root`` set.

189 - [] CI green (L2 contract + L4 regression); coverage ? floor; \$9 tracker all
?/?-resolved.

190

191 **## 12. References**

192 ****Engine code `[verified this session]`:****
``backend/main.py:12-17,42,52,307-328,331-340,342-396,638``;
``backend/api/uploads.py:64,89,123,147``; ``backend/api/job_worker.py:37``;
``backend/api/static/app.js:88,344,373``;
``backend/storage/{base.py,local.py:33,ref.py,__init__.py:43}``;
``backend/config/models.py:129,183,298``;
``backend/pipeline/segmenters/{trajectory_hybrid.py:57-63,76-85, fusion_hybrid.py:76-86,`
`gemini.py:470-485, motion.py:191-198}``; ``backend/pipeline/detectors/{native_wasb_runner.py,`
`stub_runner.py}``; ``backend/worker/__main__.py:88,98,149``; ``config.json`` (``default_segmenter:`
`gemini``). **\*\*Engine docs:\*\*** ``HOSTING_PLAN.md:42,102-108``; ``VIDEO_LOCALITY_MODEL.md``;

```
`DESKTOP_APP_PLAN.md`; `SETUP_NEW_MACHINE.md`; `PLATFORM_ARCHITECTURE.md`;
`DECOUPLED_COMPUTE/{README,02,04,07,HANDOFF}.md`; `UI_PROPOSAL.md:41-66`;
`PROJECT_DOC_TEMPLATE.md`. **This repo:** `PER_RALLY_SELECTION.md`, `REQUEST_ACCESS_FORM.md`,
`PORTABILITY.md`, `web/README.md`.
193
194 ---
195 ### Changelog
196 - 2026-06-20 ? Initial `DRAFT`. Consolidated three single-box-appliance designs
(FastAPI-native / unified-origin / decoupled-profile) into the reverse-proxied unified-origin
spine + compute-as-a-setting framing, via a multi-agent design pass; every engine-fact anchor
re-verified against source before writing (catching the native-WASB partial-wiring +
`ai_max_workers` default nuances). Folded in the owner's ingest-menu (Drive/bucket/local),
multi-provider GPU, and first-class Observability directives.
197
```

4. user (2026-06-22T09:32:50.457Z)

```
50b2bd7 chore(repo): ignore local runtime artifacts + correct stale NEXT_STEPS claims (#94)
e2570e4 docs: archive the Friendly & Multilingual tracker (engineering complete) (#93)
0bef61e docs(friendly-multilingual): close out the chapter ? log #89/#90/#91 + strengthen fitNav
tests (#92)
9f77ddc test(site): browser layout-regression CI guard + auto-collapse the nav to fit every
locale (#91)
4415fe8 feat(site): localize remaining home sections (6 langs) + smooth the responsive header
(#90)
26c9da1 feat(site): localize the hero demo mockup + #how feature block (6 languages) (#89)
7ae9cee feat(web): P2c first-run setup wizard (3 steps from /api/capabilities, 6 languages)
(#88)
cf59fbf feat(site): add es/da/id to the marketing site (parity with the SPA's 6 languages) (#87)
17242a9 feat(site): localize own-model body content to kn/hi (#86)
7b04f02 deploy(kit): align with the live-verified droplet deploy (+ the gemini-default fix)
(#85)
5e5a2ac docs(friendly-multilingual): close out the chapter ($9 DoD + UI/UX + B7 deferred) (#84)
4c0f35c fix(site): UI/UX ? compact language switcher + header overflow fix + localized hero
(#83)
0fae364 feat(web+site): A2 ? localized WhatsApp share for the reel (SPA + site) (#82)
1e52cfd deploy(droplet): tailored alpha kit for the CPU droplet (engine api. side) (#81)
64076a5 test(web): add a coverage floor (CI-gated) + i18n regression tests for the #77 audit
(#80)
8328721 docs(tracker): refresh the $7 milestone prose to merged reality (A6/A7/#77 landed) (#79)
3f0cdb8 docs(deploy): fix two would-break-the-deploy gaps in the alpha runbook
(RALLY_ALLOWED_HOSTS + gs:// jail) (#78)
9f1ec9c fix(i18n): native-review pass on the kn/hi SPA console catalogs (#77)
9a2d7d2 feat(site): A7 ? camera-placement diagram on the capture page (#76)
aa8d5e9 feat(site): A6 ? plain-language Request-Access GPU/storage questions (#75)
1cd84af feat(site): A3 ? localize recording-education + marketing to kn/hi, Noto on the site
(#74)
c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process
(analytics) (#73)
1b1fa7b feat(web+site): B3 ? trilingual marketing site via shared t()/data-i18n shim (D-CATALOG-
HOME) (#70)
f7e2bd5 chore(site): add robots.txt + a "crawling / bot protection" runbook for khelsutra.guru
(#71)
bea8a2a docs(appliance): flip T11 retention sweep to ? ? tool shipped (engine #299), scheduling
tracked (#301) (#69)
```

5. assistant (2026-06-22T09:32:54.095Z)

This is a khelsutra repo doc, but it heavily references the engine repo. Let me verify the key claims. First, let me check the repo structure and the cited khelsutra files, then check the engine references where possible.

6. user (2026-06-22T09:32:56.245Z)

deploy/DEPLOY_ALPHA.md
deploy/cloudflared.config.example.yml
deploy/droplet/README.md
deploy/droplet/config.json.example
deploy/droplet/engine.env.example
deploy/droplet/khelsutra-engine.service
docs/ANNOTATE_AND_TRAIN.md
docs/DRY_RUN_CUJ.md
docs/LOCAL_APPLIANCE_ARCHITECTURE.md
docs/MULTILINGUAL_EXPANSION.md
docs/PER_RALLY_SELECTION.md
docs/PORTABILITY.md
docs/README.md
docs/REQUEST_ACCESS_FORM.md
docs/archives/FRIENDLY_AND_MULTILINGUAL.md
web/.gitignore
web/README.md
web/index.html
web/package-lock.json
web/package.json
web/src/App.tsx
web/src/api/client.test.ts
web/src/api/client.ts
web/src/api/locale.test.ts
web/src/api/types.ts
web/src/api/upload.test.ts
web/src/components/CompileResult.tsx
web/src/components/IngestPanel.test.tsx
web/src/components/IngestPanel.tsx
web/src/components/LanguageSwitcher.tsx
web/src/components/QueryBar.examples.test.ts
web/src/components/QueryBar.test.tsx
web/src/components/QueryBar.tsx
web/src/components/RallyGrid.tsx
web/src/components/SelectionFooter.test.tsx
web/src/components/SelectionFooter.tsx
web/src/components/SetupWizard.test.tsx
web/src/components/SetupWizard.tsx
web/src/components/StatusChip.tsx
web/src/components/TapFilters.test.tsx
web/src/components/TapFilters.tsx
web/src/components/a4.touch.test.tsx
web/src/components/a8.status.test.tsx
web/src/components/build-error.test.tsx
web/src/components/upload.test.tsx
web/src/hooks/useIngestJob.test.ts
web/src/hooks/useIngestJob.ts
web/src/hooks/useRallySelection.ts
web/src/hooks/useUpload.test.ts
web/src/hooks/useUpload.ts
web/src/i18n/catalog-invariants.test.ts
web/src/i18n/hygiene.test.ts
web/src/i18n/i18n.test.tsx
web/src/i18n/index.ts
web/src/index.css
web/src/lib/errorText.test.ts
web/src/lib/errorText.ts
web/src/lib/errors.test.ts
web/src/lib/errors.ts
web/src/lib/floor.test.ts
web/src/lib/floor.ts
web/src/lib/format.test.ts
web/src/lib/format.ts
web/src/lib/log.ts
web/src/lib/query.test.ts

```
web/src/lib/query.ts
web/src/lib/selection.test.ts
web/src/lib/selection.ts
web/src/main.tsx
web/src/test/setup.ts
web/src/vite-env.d.ts
web/tsconfig.json
web/vite.config.ts
```

7. user (2026-06-22T09:32:57.327Z)

```
ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
8bbc557 feat(web): VITE_API_BASE ? let the SPA target a split api.* origin (default same-origin /api) (#64)
4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280) (#55)
```

8. assistant (2026-06-22T09:33:01.807Z)

The doc references PRs that exist. Note the doc says `web/` is a "React SPA" / "React scaffold" but the files are `.tsx`. Let me verify some of the doc's khelsutra-specific claims and check which PRs the doc references (the doc references both engine PRs and khelsutra PRs). Let me check key claims.

9. user (2026-06-22T09:33:03.924Z)

```
{
  "name": "khelsutra-web",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "description": "KhelSutra Studio ? operator console SPA (drives the rally indexer engine).",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "preview": "vite preview",
    "typecheck": "tsc --noEmit",
    "test": "vitest run",
    "test:coverage": "vitest run --coverage"
  },
  "dependencies": {
    "@fontsource-variable/noto-sans-devanagari": "^5.2.8",
    "@fontsource-variable/noto-sans-kannada": "^5.2.9",
    "i18next": "^26.3.1",
    "i18next-browser-languagedetector": "^8.2.1",
    "i18next-icu": "^2.4.3",
    "react": "^19.0.0",
    "react-dom": "^19.0.0",
    "react-i18next": "^17.0.8"
  },
  "devDependencies": {
    "@tailwindcss/vite": "^4.0.0",
    "@testing-library/dom": "^10.4.1",
    "@testing-library/react": "^16.3.2",
    "@types/node": "^26.0.0",
    "@types/react": "^19.0.0",
    "@types/react-dom": "^19.0.0",
    "@vitejs/plugin-react": "^4.3.4",
    "@vitest/coverage-v8": "^3.2.6",
    "jsdom": "^29.1.1",
    "tailwindcss": "^4.0.0",
    "typescript": "^5.7.0",
    "vite": "^6.0.0",
```

```

    "vitest": "^3.0.0"
  }
}
---SetupWizard exists?---
web/src/components/SetupWizard.tsx

```

10. user (2026-06-22T09:33:05.641Z)

```

=== DEPLOY_ALPHA exists ===
deploy/DEPLOY_ALPHA.md
deploy/cloudflared.config.example.yml
=== DRY_RUN_CUJ section 9 ===
15:- **HTTP bucket-ingest ALREADY EXISTS in the engine (shipped M2 / [#280]) `[verified
file:line]`.** `POST /api/process` accepts a `gs://`/`gcs://` (also `s3://`/`local://`)
`video_path` directly. Submit resolves the URI and **skips the local-existence check for non-
local backends** (`backend/main.py:274,283` ? `storage.resolve_input_ref`; `gs://`/`gcs`
normalized in `backend/storage/ref.py:60`); the job worker then fetches it to scratch
(`_execute_processing` ? `storage.acquire_local_input`, `main.py:159`) and runs the **same**
hash ? validate ? segment ? DB pipeline as a local file ? writing a queryable `video_id` into
the API's SQLite `videos`/`play_segments`. Proven end-to-end by
`tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches` (gs:// ? 202,
not 404 ? job `done` ? `db.get_video(video_id)` row with the **source URI preserved**), plus the
hosted-mode-rejection / unknown-scheme-400 / CLI-variant tests.
21:**Path B ? Full "point at a bucket video IN the UI" (SPA-only ? smaller than first scoped).**
The **HTTP bucket-ingest endpoint already exists** (§1) ? so Path B is **only the SPA front-
half** (appliance S1/S2): an **ingest** input (enter a `gs://` URI) ? `POST /api/process` ? an
**ingest + progress** screen polling `GET /api/jobs/{id}` ? on `done`, load `result.video_id`
(which the existing back-half already renders). **No net-new engine endpoint** ? B-P1 is already
done; the remaining work is **B-P2 (the SPA)****.
26: `gsutil cp gs://<bucket>/<match>.mp4 ./incoming/` (or the worker fetch), then `POST
/api/process {"video_path":"<abs local path>","segmenter":"gemini"}` ? poll `GET
/api/jobs/{job_id}` until `result.video_id`. *(`gemini` needs `GEMINI_API_KEY`; `motion` is key-
free but hides fabricated metadata.)*
34:1. **Scope:** Path A (cheapest dry run now) or Path B (the in-UI bucket ingest ? **SPA-
only**); the engine endpoint already accepts `gs://`, see §1)? *Recommend A first* (proves the
multilingual workflow fast), then B as the appliance build. ? ? **RESOLVED: A first (done, §7),
then B (B-P2, the SPA ingest/progress screen).**
37:4. **Which bucket video:** e.g. `gs://khelsutra-rally-
corpus/videos/mahadevpura_smoke_180s.mp4` (already staged, known-good) or another clean ?720p
clip (must pass the engine validators).
49:- [x] *(Path B = B-P2, SPA-only)* the bucket video is pointed-at **from the UI** (no CLI pre-
step). ? ? **DONE (live e2e, 2026-06-21), see §9.** In the SPA: entered a `gs://` URI ? `POST
/api/process` ? polled `/api/jobs` ? **12 rallies loaded** ? Build ? **reel downloaded (1080p,
76s)** , all from the UI. Surfaced + fixed an engine compile bug along the way (engine
[#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289)). **Screencast:**
`gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`.
54:1. `gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4` (1080p/30fps) ? `gsutil
cp` local ? `POST /api/process {segmenter: gemini}`.
60:Screencast/proof: per-locale screenshots; both reels on the local machine. Screencast of the
i18n SPA is at `gs://khelsutra-rally-corpus/promo/khelsutra-i18n-screencast.mp4`.
68:**What the code actually does `[verified file:line, this session]`:** `POST /api/process` has
accepted bucket/Drive URIs since **engine M2 ([#280](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/280))**. A `gs://`/`gcs://`/`s3://`/`local://` `video_path`:
69:1. is resolved at submit (`backend/main.py:274` ? `storage.resolve_input_ref`; `gs://`/`gcs`
at `backend/storage/ref.py:60`; explicit scheme wins under default `input_backend=local` at
`ref.py:87`),
80:## 9. ? Run result ? Path B, live in-UI bucket?reel CUJ (2026-06-21)
82:1. Opened the SPA ? the **"Process a video"** ingest panel (B-P2) ? pasted `gs://khelsutra-
rally-corpus/videos/mahadevpura_smoke_180s.mp4` ? **Process video**.
85:4. **Screencast of the whole CUJ** (Playwright-recorded; the ~3-min Gemini wait fast-
forwarded): `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`.
88:`/api/compile` **500'd** on a bucket-ingested video: `videos.filepath` keeps the **source
URI** (`gs://`/`, M2 provenance), and the stitcher was handed that directly ? ffmpeg can't open a
`gs://` path. **Fix:** compile now resolves the source via `storage.acquire_local_input` first
(identity for a local path; fetch-to-scratch for a bucket URI) + a regression test ? engine

```

92: - `2026-06-21` ? **\$8 correction:** verified B-P1 (`gs://`?`/api/process`?API-DB `video\_id`) is already shipped + tested (M2/#280); \$1/\$2/\$4/\$6 reframed so Path B = SPA-only (B-P2). Sibling stale claims fixed in `LOCAL\_APPLIANCE\_ARCHITECTURE.md` (`/api/health` exists; bucket-URI ingest not a net-new endpoint; T12 engine-half done) + `README.md`.

93: - `2026-06-21` ? **\$9: Path-B live e2e DONE** ? the in-UI `gs://`?reel CUJ ran end to end (12 rallies ? reel downloaded); surfaced + fixed the compile-from-bucket 500 (engine #289); screencast uploaded to the promo bucket. **DoD \$6 box 3 ?.**

11. assistant (2026-06-22T09:33:10.124Z)

The DRY_RUN_CUJ doc cross-references confirm T12. Now the key issue: the doc references PR `[#11]` (D4) and `[#66]` (T4) and `[#56]` (T12) and `[#64]` (alpha deploy). Let me verify these are real khelsutra PRs and check the T4 PR number specifically (the doc says `#66` for local upload). Let me also verify the engine-side claims that I can verify by checking if the engine repo is accessible locally.

12. user (2026-06-22T09:33:11.794Z)

badminton-highlight-indexer
=== engine repo present ===
689c738 docs(pkg): mark P2c ? + the loop closed ? v1 LIGHT is feature-complete (#313)
293e5c0 feat(pkg): re-bundle backend/ui ? serve the P2c first-run wizard from indexer --app (#312)
f8b2c3b feat(pkg): P2b-install ? batteries-included install + truly-local 'motion' first-run default (gate 08) (#310)

13. user (2026-06-22T09:33:13.732Z)

=== PR #11 (RallyPicker)? ===
05915f2 docs(next-steps): per-rally MVP/M0 COMPLETE (P1?P5, #28/#30/#31); CUJ screencast + M1?M5 next (#32)
8bf200d feat(web): per-rally MVP P5 ? load real rallies + Build?compile?preview?download (MVP complete) (#31)
aff8de8 feat(web): per-rally MVP P4 ? SelectionFooter + honest min_shot_count warning (#30)
66805ab docs(next-steps): per-rally P3 (parseQuery query bar) merged (#28); P4 next (#29)
d537d19 feat(web): per-rally MVP P3 ? deterministic parseQuery() query bar (traced + adversarially hardened) (#28)
62598e7 feat(web): per-rally MVP P2 ? selectable rally grid + selection hook (unit-tested) (#21)
aac7e0c feat(web): per-rally MVP P1 ? SPA scaffold + typed engine API client (#14)
cb74ce5 docs(web): tracked design for interactive per-rally selection (RallyPicker) (#11)
=== count PRs (max merge #) ===
#93
#94
#280
#299
#301

14. assistant (2026-06-22T09:33:17.502Z)

The engine repo IS locally present. PR #11 is confirmed as the RallyPicker design doc. The doc references PRs in two repos ? khelsutra (#11, #56, #64, #66) and engine (#280, #299, #301). Let me verify the critical engine-side claims now since the engine repo is present. Let me check the engine's master branch is synced and verify the key falsifiable engine claims.

15. user (2026-06-22T09:33:20.460Z)

work/golden-corpus-hygiene
=== main.py:638 host bind ===
agg = db_instance.get_video_cost(video_id)
fx = global_config.billing.fx_inr_per_usd
agg["total_cost_inr"] = billing.to_inr(agg["total_cost_usd"], fx)

```

    agg["fx_inr_per_usd"] = fx
    return agg

@app.get("/api/jobs/{job_id}")
def get_job(job_id: str):
    """Job state: {status, progress, result, reports}. `status` = execution state
    (queued|running|done|failed); the pipeline verdict is result["status"]."""
    job = db_instance.get_job(job_id)
    if not job:
        raise HTTPException(status_code=404, detail=f"Unknown job_id '{job_id}'")
    result = job.get("result")
    return {
=== main.py:42 CORS ===
from fastapi.responses import FileResponse, HTMLResponse, JSONResponse
from backend.pipeline.validators.base import registry
from backend.infrastructure.database import Database
from backend.pipeline.segmenters.base import segmenter_registry
from backend.pipeline.stitcher.compiler import VideoCompiler
from backend.pipeline.stitcher.watermark_i18n import localize_watermark
from backend.utils.hashing import compute_video_id # canonical file-identity hashing
from backend.utils.paths import resolve_under_root, safe_output_filename, validate_input_path
from backend.utils.redact import redact_secrets # P0b: mask secret-shaped fields in /api/config
from backend.utils.single_instance import acquire_lock # P0e: one instance per database
from backend.utils.ffmpeg import ffmpeg_bin, ffprobe_bin # P2a: single ffmpeg/ffprobe resolver
from backend.config import ( # new typed config
    load_config as load_app_config,
    write_light_default_config, # P2b: batteries-included first-run config seeding
    AppConfig,
    StitchingConfig,
    enforce_startup_checks,
    StartupCheckError,

```

16. assistant (2026-06-22T09:33:26.506Z)

The engine repo is on a different branch (`work/golden-corpus-hygiene`) and the line numbers have drifted significantly ? main.py:638 is no longer the host bind, and main.py:42 is no longer the CORS line. This is expected: the doc anchors line numbers that drift as the engine evolves. The doc itself flags engine-side line numbers as a moving target. The important thing per my task is verifying the khelsutra-side claims and the high-level status claims, not chasing drifted engine line numbers (the engine is on a feature branch, not master, so line numbers aren't even authoritative).

Let me verify the engine claims at the semantic level (do these features exist?) rather than exact line numbers, and confirm the key status claims. Let me check the CORS, host bind, and /api/health existence semantically.

17. user (2026-06-22T09:33:28.278Z)

```

76:# CORS for the web UI. allow_origins='' with allow_credentials=True is rejected by browsers
per the
82:    allow_origins=cors_origins_from_env(),
103:    return hosts or ["localhost", "127.0.0.1", "testserver"]
129:    return "127.0.0.1"
307:@app.get("/api/health")
308:def health():
340:@app.get("/api/capabilities")
1082:    bind_host = "127.0.0.1"
1098:    if bind_host not in ("127.0.0.1", "localhost", ":::1"):

```

18. assistant (2026-06-22T09:33:34.147Z)

Important finding. The engine's CORS is NO LONGER `allow\_origins=''`. Line 76 says: "allow_origins='' with allow_credentials=True is rejected by browsers" and line 82 uses

`\_cors\_origins\_from\_env()` ? meaning CORS is now configurable/restricted, not hardcoded `\*`. Also `/api/capabilities` NOW EXISTS at line 340 (the doc says it's "Absent (net-new)" and "don't exist"). Let me dig into these.

19. user (2026-06-22T09:33:34.621Z)

```
72         origins = [o.strip() for o in raw.split(",") if o.strip()]
73         return origins or ["*"]
74
75
76     # CORS for the web UI. allow_origins='*' with allow_credentials=True is rejected by
browsers per the
77     # fetch spec + is unsafe to carry into multi-tenant; this tool uses no cookies, so
credentials stay
78     # off. The origin list is configurable via the env var above (default ["*"]) so a hosted
deploy can
79     # lock it to its origin; the middleware is fixed at startup.
80     app.add_middleware(
81         CORSMiddleware,
82         allow_origins=_cors_origins_from_env(),
83         allow_credentials=False,
84         allow_methods=["*"],
85         allow_headers=["*"],
86     )
87
88
89     def _allowed_hosts_from_env(env: Optional[Dict[str, str]] = None) -> List[str]:
90         """P0c: Host-header allowlist ? a DNS-rebinding guard for the localhost appliance. A
malicious
91         web page can point a domain it controls at 127.0.0.1 and script requests to the
local server;
92         the browser sends `Host: attacker.com`, which this rejects (the server only
answers loopback
93         Host headers by default). `RALLY_ALLOWED_HOSTS` (comma-separated) overrides; the
default keeps
94         Starlette's TestClient host `testserver` so the in-process suite isn't broken.
Read from the
95         ENV (not config.json) so importing `backend.main` does zero filesystem I/O; a
routable/edge
96         deploy sets its public host (or ``*`` when fronted by an auth gate that validates
the host)."""
97         src = os.environ if env is None else env
98         raw = (src.get("RALLY_ALLOWED_HOSTS", "") or "").strip()
99         hosts = [h.strip() for h in raw.split(",") if h.strip()]
100         # IPv4 loopback (the default bind) + the TestClient host. NB: Starlette's
TrustedHostMiddleware
101         # matches on host.split(":")[0], which mangles a bracketed IPv6 Host (":::1:port"),
so IPv6
102         # loopback is NOT supported by this guard ? access the appliance via
127.0.0.1/localhost.
103         return hosts or ["localhost", "127.0.0.1", "testserver"]
104
105
106     # DNS-rebinding guard. Bound to localhost by default, so only loopback Host headers are
honoured
107     # (a remote page can't read responses cross-origin AND its Host header is rejected
here). www_redirect
108     # off so an unexpected `www.` host is rejected rather than 307-redirected.
109     app.add_middleware(
110         TrustedHostMiddleware,
111         allowed_hosts=_allowed_hosts_from_env(),
```

20. assistant (2026-06-22T09:33:38.966Z)

Confirmed: the engine's CORS default is still `["\*"]` (line 73: `return origins or ["\*"]`), but

it is now **configurable** and `allow_credentials=False`. The doc's flat claim "CORS ````" is partially stale ? it's still default-```` but now env-lockable. More importantly, the doc's structural-safety framing is now augmented by added defenses (TrustedHostMiddleware DNS-rebinding guard, exposure-safety boot posture). Let me verify `/api/capabilities` and `/api/health` content, since the doc lists capabilities as "Absent (net-new)".

21. user (2026-06-22T09:33:39.915Z)

```
307 @app.get("/api/health")
308 def health():
309     """Cheap liveness/readiness probe (no auth) for the SPA's offline banner
310     (HOSTING_PLAN.md:42). Reports the configured sport + default segmenter so the UI can
311     confirm the engine is reachable and which detector is wired. Never raises."""
312     sport = getattr(global_config, "sport", None)
313     indexing = getattr(global_config, "indexing", None)
314     return {
315         "status": "ok",
316         "sport": sport,
317         "default_segmenter": getattr(indexing, "default_segmenter", None),
318     }
319
320
321 # P1: the bundled SPA pins this ? the contract version of the /api surface. BUMP when an
existing
322 # endpoint's request/response shape changes incompatibly (purely additive endpoints need
no bump).
323 API_VERSION = 1
324
325
326 def _engine_version() -> str:
327     """Installed package version (wheel/editable); 'unknown' if not importable as a
distribution."""
328     try:
329         return _pkg_version("badminton-highlight-indexer")
330     except PackageNotFoundError:
331         return "unknown"
332
333
334 @app.get("/api/version")
335 def api_version():
336     """Version handshake so a bundled SPA can detect engine skew (PACKAGING_PLAN.md
P1)."""
337     return {"engine_version": _engine_version(), "api_version": API_VERSION, "ui":
_bundled_ui_version()}
338
339
340 @app.get("/api/capabilities")
341 def capabilities():
342     """Honest probe of what THIS box can do ? the first-run wizard renders the compute
step from
343     it (PACKAGING_PLAN.md P1). No secrets: reports only whether a Gemini key is PRESENT,
never its
344     value. Best-effort + never raises (the wizard degrades gracefully)."""
345     indexing = getattr(global_config, "indexing", None)
346     deployment = getattr(global_config, "deployment", None)
```

22. assistant (2026-06-22T09:33:46.992Z)

This is a major staleness finding. `/api/capabilities` and `/api/health` **now exist** in the engine. The doc:

- §3 "Absent (net-new)" lists `GET /api/capabilities` as absent ? **STALE** (it exists at `main.py:340`).
- §3 says `/api/health` "now EXISTS" ? that one is correct.
- §4.6 Reuse map lists `/api/capabilities`, `/api/health` under "Net-new" ? **STALE** for capabilities

- \$9 tracker T7 (capabilities + health + banner) marked ? Todo ? ****partially stale**** (engine half shipped).
- \$11 DoD: `/api/capabilities` + `/api/health` + banner ? engine endpoints now exist.

Note: the engine is on branch `work/golden-corpus-hygiene`, not master. Let me confirm these endpoints are on the engine's master (not just this feature branch) to be precise about the verdict.

23. user (2026-06-22T09:33:49.773Z)

```
=== capabilities on master? ===
origin/master:backend/main.py:340:@app.get("/api/capabilities")
=== health on master? ===
=== CORS on master ===
origin/master:backend/main.py:76:# CORS for the web UI. allow_origins='' with
allow_credentials=True is rejected by browsers per the
origin/master:backend/main.py:82:    allow_origins=cors_origins_from_env(),
```

24. assistant (2026-06-22T09:33:55.675Z)

Confirmed on engine `origin/master`:

- `/api/capabilities` exists (main.py:340) ? doc's "Absent (net-new)" claim is STALE.
- The `git grep /api/health` returned nothing for the decorator on its own line, but I saw it at line 307. Let me double-check (grep may have matched differently).
- CORS is now `\_cors\_origins\_from\_env()`, NOT hardcoded `` ? the doc's flat `[verified backend/main.py:42]` "CORS ``" anchor is stale (line drifted + now configurable).

Let me confirm health on master and check the native WASB status claim (the doc says native is "in-flight on branch feat/native-wasb-runner" and "trajectory_hybrid._build_native_runner still raises").

25. user (2026-06-22T09:33:59.290Z)

```
=== health endpoint on master ===
307:@app.get("/api/health")
1092:    threading.Thread(target=_open_browser_when_ready, args=(url, url +
"api/health"),
=== native wasb branch still exists? ===
    feat/native-wasb-runner
=== native_wasb_runner present on master? ===
backend/pipeline/detectors/native_wasb_runner.py
tests/test_native_wasb_runner.py
=== trajectory_hybrid._build_native_runner on master ===
56:         raise NotImplementedError
58:     def _build_native_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner,
Dict[str, Any]]:
61:         raise NotImplementedError(
73:         ``_build_native_runner``. Anything else (``"auto"``) delegates to ``_build_runner``
86:         return self._build_native_runner(idx_cfg)
133:         ``FusionSegmenter`` overrides this to apply Gate A/B when thresholds are raised.
```

26. assistant (2026-06-22T09:34:06.692Z)

Good ? confirmed on engine master:

- `/api/health` exists (main.py:307) ? doc correct on health.
- `native\_wasb\_runner.py` is now on ****master**** (not just the `feat/native-wasb-runner` branch). The doc says native WASB is "in-flight on branch `feat/native-wasb-runner`" ? this is now partially stale: the runner file has merged to master. The branch still exists.
- `trajectory\_hybrid.\_build\_native\_runner` still raises `NotImplementedError` (line 61) ? doc correct on that detail.

Now let me verify the khelsutra-side claims more precisely. The doc's \$9 tracker references khelsutra PRs #66 (T4) and #56 (T12), and the alpha deploy uses #64. The latest khelsutra PR is #94. Let me check the doc's most actionable claims: the IngestPanel (bucket ingest, T12), the

upload wiring (T4), and the deploy kit. Let me also verify the engine droplet IP claim and the `default\_segmenter: gemini` config claim.

27. user (2026-06-22T09:34:09.653Z)

```
=== IngestPanel handles gs:// + upload? ===
2:// video: UPLOAD a local file (primary ? alpha users have a file, not a URI; chunked to the
engine's
3:// /api/upload/* endpoints) or paste a `gs://` bucket URI / host path (secondary). Both
converge on
11:import { useUpload } from "../hooks/useUpload";
26: // On upload completion, feed the returned server PATH straight into the engine processing
back-half.
27: const upload = useUpload((path) => submit(path, locale));
28: const busy = ingestBusy || upload.busy;
36:   if (file) upload.start(file);
40:   upload.cancel();
44:   upload.reset();
50: // localized line. Surface whichever half failed (upload or engine) ? they're mutually
exclusive.
51: const errorState = upload.phase.status === "error" ? upload.phase.error : phase.status ===
"error" ? phase.error : null;
59:   {/* PRIMARY: upload a local file (T4). A native file input, visually a button (?44px,
A4). */}
73:     aria-label={t("ingest.uploadAria")}
82:   {/* SECONDARY: or point at a cloud bucket / host path. */}
107:    (A8). Upload shows a determinate % bar; the engine phase is indeterminate (phase
words). */}
=== VITE_API_BASE wiring ===
web/src/api/client.test.ts:27: it("defaults to same-origin /api when VITE_API_BASE is unset",
() => {
web/src/api/client.test.ts:31: it("uses VITE_API_BASE for a split deploy (SPA on Pages + api.*
tunnel)", () => {
web/src/api/client.test.ts:32:   vi.stubEnv("VITE_API_BASE", "https://api.khelsutra.guru/api");
web/src/api/client.test.ts:37:   vi.stubEnv("VITE_API_BASE",
"https://api.khelsutra.guru/api/");
web/src/api/client.ts:20: * set `VITE_API_BASE` at build time to the engine's API base
**including the `/api` path**
web/src/api/client.ts:25: return (import.meta.env.VITE_API_BASE ?? "/api").replace(/\/+$/, "");
web/src/vite-env.d.ts:9: readonly VITE_API_BASE?: string;
```

28. user (2026-06-22T09:34:11.331Z)

```
=== engine config.json default_segmenter ===
  "default_segmenter": "gemini",
=== droplet IP referenced in deploy ===
deploy/DEPLOY_ALPHA.md:31:- A **box** to run the engine: the CPU droplet `168.144.126.176`
(Gemini-only ? no GPU; fine for alpha) **or** your Windows+WSL GPU box (real WASB). `ffmpeg` +
Python 3.8+ installed; the `badminton-highlight-indexer` engine checked out.
deploy/droplet/README.md:4:(`168.144.126.176`, Gemini, **no GPU**). Values are pre-baked for
**`app.khelsutra.guru` /
deploy/droplet/README.md:87:curl --max-time 5 http://168.144.126.176:8000/api/health      # ->
refused / timeout
deploy/droplet/README.md:100:> **? Live-verified on the droplet (2026-06-22).** Engine installed
+ booted on `168.144.126.176`; full
docs/DRY_RUN_CUJ.md:36:3. **Engine host:** local Windows (simplest, Gemini/motion) . the CPU
droplet `168.144.126.176` (shared testbed ? careful) . a GPU VM (only for WASB).
docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48:- **Droplet reality:** `168.144.126.176`, **CPU-only, NO
GPU**, region `blr1`; co-hosts the marketing site (`/srv/khelsutra-site`, node `:8787`) **and**
the engine (`~/rally`); ran the full train+infer e2e against DO Spaces with the **GPU mocked**
`[SETUP_NEW_MACHINE.md, DECOUPLED_COMPUTE/HANDOFF.md]`. Shared testbed: MinIO `9000/9001` + GCS
emulator `9023` also run there.
```

29. assistant (2026-06-22T09:34:20.920Z)

Confirmed:

- IngestPanel handles both upload (T4) and `gs://` bucket URI (T12) ? both shipped. The doc's tracker marks these correctly (T4 ?, T12 ?).
- VITE_API_BASE wiring (#64) confirmed for split deploy.
- Engine `config.json` `default\_segmenter: gemini` ? doc's §12 reference correct.
- Droplet IP `168.144.126.176`, blr1, CPU-only ? consistent with deploy docs.

Now, the most significant staleness theme: the alpha deploy. The doc's §9 says alpha stand-up is the "owner step" (not done). But the MEMORY context says T2 alpha deploy is "DONE + LIVE (2026-06-22): app.khelsutra.guru live". And the engine has shipped M9-auth, exposure-safety. The droplet README says "Live-verified on the droplet (2026-06-22)". Let me check whether the doc's §9 "actual stand-up ... is the owner step" is now stale (alpha is live). Let me check the deploy/droplet README and the alpha status.

30. user (2026-06-22T09:34:21.746Z)

```
95
96      **The shareable CUJ:** open `app.khelsutra.guru` ? sign in (email OTP) ? **upload a
video** (lands in the
97      jail) ? Process ? pick rallies ? Build ? download the reel. *(Pasting a `gs://` URI is
rejected by the
98      jail on the exposed box ? upload is the alpha path; see `../DEPLOY_ALPHA.md` §7.)*
99
100     > **? Live-verified on the droplet (2026-06-22).** Engine installed + booted on
`168.144.126.176`; full
101     > suite **1313 passed** + ruff/mypy clean on the box; boot-posture fail-fast, the Host-
guard `400?200`,
102     > the path-jail (`gs://`/etc/passwd`/`...` ? 400), the T4 upload?jail, and the edge-
auth `401`-gate all
103     > exercised live. Tunnel + Access + Pages are up; the full **DNS ? Cloudflare ? Access ?
tunnel ? engine**
104     > chain is confirmed (`/api/health` = 200 through an Access session; unauthenticated ?
302 to the login).
105     > Two things still need a human: the rotated **Gemini key** (`/etc/khelsutra/engine.env`
? restart) and a
106     > **browser one-time-PIN login** to exercise the upload?reel CUJ.
107     >
108     > **? Two gotchas the live run surfaced:**
109     > - **`indexing.default_segmenter` MUST be `gemini` on a CPU box** (set in
`config.json.example`). The SPA
110     > sends no segmenter, so the engine uses its default ? the shipped default
`yolo_hybrid` needs torch/GPU
111     > and every job fails `No module named 'torch'` without this.
112     > - **Without a valid `GEMINI_API_KEY`, a job finishes `done` with 0 rallies** ? the
missing-key error lands
113     > only in the engine log, not the job status. Set the key before sharing.
114
```

31. assistant (2026-06-22T09:34:32.721Z)

This confirms a significant staleness theme. The alpha deploy is **LIVE-verified on 2026-06-22** (`app.khelsutra.guru` up, full DNS?CF?Access?tunnel?engine chain confirmed, suite 1313 passed). The doc (last updated 2026-06-21) says in §9: "the actual stand-up (tunnel + Pages + CF Access + key rotation) is the **owner** step" ? this is now largely done (only Gemini key rotation + browser OTP remain). And §11 DoD box 1 (CUJ-1 e2e on droplet behind gate) and box 2 (raw :8000 refused, tested) ? the droplet README says the posture test was "exercised live" ? so the doc's DoD checkboxes are stale-pending.

Let me check the doc's §3 droplet claim about marketing site (`/srv/khelsutra-site`, node `:8787`) vs the deploy reality, and confirm the doc's status header. The doc is `DRAFT`, last updated 2026-06-21, while much has shipped. Let me check docs/README.md for how this doc is indexed and whether DESKTOP_APP_PLAN supersession holds.

32. user (2026-06-22T09:34:36.120Z)

<persisted-output>

Output too large (140.6KB). Full output saved to: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\tool-results\ba2pjspt.txt

Preview (first 2KB):

=== how README indexes this doc ===

8: bugs the runs surfaced are fixed (#281, #289). Was a milestone of

`LOCAL\_APPLIANCE\_ARCHITECTURE.md`.

19:- **`LOCAL\_APPLIANCE\_ARCHITECTURE.md`** ? the single-box, UI-powered rally indexer + reel

=== site server port :8787 / khelsutra-site ===

```
site/node_modules/.package-lock.json:2:  "name": "khelsutra-site",
site/node_modules/tr46/lib/mappingTable.json:1:[[0,44],2],[45,46],2],[47,2],[48,57],2],[58,64],2],[65,1,"a"],[66,1,"b"],[67,1,"c"],[68,1,"d"],[69,1,"e"],[70,1,"f"],[71,1,"g"],[72,1,"h"],[73,1,"i"],[74,1,"j"],[75,1,"k"],[76,1,"l"],[77,1,"m"],[78,1,"n"],[79,1,"o"],[80,1,"p"],[81,1,"q"],[82,1,"r"],[83,1,"s"],[84,1,"t"],[85,1,"u"],[86,1,"v"],[87,1,"w"],[88,1,"x"],[89,1,"y"],[90,1,"z"],[[91,96],2],[[97,122],2],[[123,127],2],[[128,159],3],[160,1," "],[[161,167],2],[168,1,"?"],[169,2],[170,1,"a"],[[171,172],2],[173,7],[174,2],[175,1,"?"],[[176,177],2],[178,1,"2"],[179,1,"3"],[180,1,"?"],[181,1,"?"],[182,2],[183,2],[184,1,"?"],[185,1,"1"],[186,1,"o"],[187,2],[188,1,"1?4"],[189,1,"1?2"],[190,1,"3?4"],[191,2],[192,1,"à"],[193,1,"á"],[194,1,"â"],[195,1,"ã"],[196,1,"ä"],[197,1,"å"],[198,1,"æ"],[199,1,"ç"],[200,1,"è"],[201,1,"é"],[202,1,"ê"],[203,1,"ë"],[204,1,"ì"],[205,1,"í"],[206,1,"î"],[207,1,"ï"],[208,1,"ð"],[209,1,"ñ"],[210,1,"ò"],[211,1,"ó"],[212,1,"ô"],[213,1,"ö"],[214,1,"ø"],[215,2],[216,1,"ø"],[217,1,"ù"],[218,1,"ú"],[219,1,"û"],[220,1,"ü"],[221,1,"ý"],[222,1,"þ"],[223,6,"ss"],[[224,246],2],[247,2],[[248,255],2],[256,1,"?"],[257,2],[258,1,"?"],[259,2],[260,1,"?"],[261,2],[262,1,"?"],[263,2],[264,1,"?"],[265,2],[266,1,"?"],[267,2],[268,1,"?"],[269,2],[270,1,"?"],[271,2],[272,1,"?"],[273,2],[274,1,"?"],[275,2],[276,1,"?"],[277,2],[278,1,"?"],[279,2],[280,1,"?"],[281,2],[282,1,"?"],[283,2],[284,1,"?"],[285,2],[286,1,"?"],[287,2],[288,1,"?"],[289,2],[290,1,"?"],[291,2],[292,1,"?"],[293,2],[294,1,"?"],[295,2],[296,1,"?"],[297,2],[298,1,"?"],[299,2],[300,1,"?"],[301,2],[302,1,"?"],[303,2],[304,1,"i?"]],[305,2],[[306,307],1,"ij"]
...

```

</persisted-output>

33. assistant (2026-06-22T09:34:46.584Z)

The site dir exists (`khelsutra-site` in node_modules). The README indexes the doc. The grep hit node_modules noise. The doc's \$3 droplet detail (`/srv/khelsutra-site`, node `:8787`) is an engine-side runbook claim I can't cleanly verify from here, but it's a minor operational detail.

I have enough to finalize. Let me do one final check on the doc's \$0/\$3 CORS claim severity ? whether the doc's central safety thesis ("zero app-layer auth and CORS `\*`") is materially stale given the engine added DNS-rebinding guards, host allowlist, exposure-safety boot posture, and M9 edge-auth. Let me confirm the exposure-safety boot posture exists (the doc's \$9 mentions "engine #294").

34. user (2026-06-22T09:34:48.708Z)

=== exposure-safety / enforce_startup_checks on engine master ===

origin/master:backend/api/auth.py

origin/master:backend/config/__init__.py

origin/master:backend/config/startup.py

origin/master:backend/main.py

origin/master:backend/worker/__main__.py

=== M9 auth / Cf-Access on master ===

origin/master:backend/api/job_worker.py

origin/master:backend/config/models.py

origin/master:backend/infrastructure/database.py

origin/master:backend/main.py

35. assistant (2026-06-22T09:34:54.644Z)

Confirmed. The engine has shipped substantial auth/exposure hardening (`backend/api/auth.py`, `enforce\_startup\_checks`, M9 `Cf-Access-Authenticated-User-Email` + `owner\_id`) since the doc was written. The doc's central thesis ("zero app-layer auth") is now materially stale ? there IS

now an edge-auth integration layer (`backend/api/auth.py`) plus exposure-safety boot posture.

I have thorough evidence. Let me compile findings. Summary of the doc's status: it's a `DRAFT`/`IN PROGRESS` tracker (§9 source of truth) whose tracked work has substantially shipped since 2026-06-21, and several engine-fact anchors are stale because the engine moved.

36. user (2026-06-22T09:35:36.754Z)

Structured output provided successfully

37. assistant (2026-06-22T09:35:48.043Z)

I've completed the audit of `docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md`. The structured output has been returned.

Key findings: the doc is a still-active DRAFT tracker (last updated 2026-06-21) with ****significant staleness**** concentrated in engine-fact anchors and §9 progress, while its khelsutra-side claims hold up:

****High-severity (a reader would act on these):****

- `/api/capabilities` is listed as "Absent (net-new)" but now exists on engine `origin/master` at `backend/main.py:340`.
- The central safety thesis "zero app-layer auth and CORS ``" is materially stale ? the engine now has env-configurable CORS (`allow\_credentials=False`), a TrustedHost DNS-rebinding guard, `backend/api/auth.py` edge-auth, and exposure-safety boot posture.
- The alpha deploy is described as an unfinished "owner step," but `deploy/droplet/README.md:100` records it LIVE-verified 2026-06-22 (`app.khelsutra.guru`, full DNS?CF?Access?tunnel?engine chain, posture test exercised).

****Medium:**** native_wasb_runner.py has merged to engine master (doc says "in-flight on branch"); the stale "last updated" date.

****Correct:**** PR references #11/#56/#64/#66, IngestPanel upload+gs:// handling, VITE_API_BASE wiring, T12 live CUJ.

****Unverifiable from this repo:**** engine GitHub PR #299/#301 retention claims, live-droplet filesystem paths/ports, and the DESKTOP_APP_PLAN supersession (an engine-repo doc).

Not yet an archive candidate ? open owner gates remain (edge-auth choice, retention, cost ceiling, GPU host, P5 bridge) ? but it needs a refresh pass to re-sync the engine facts and §9 tracker.